

Week 5 Discussion: Duration & Forward Pricing (Lec 11–13)

Main focus: (i) bond pricing from yield and **duration** as price sensitivity; (ii) **forward rates** from spot rates; (iii) **forward pricing** via static no-arbitrage. All rates are continuously compounded.

1 Part 1: Bond Duration & Sensitivity

Module A) Duration (Macaulay w/ continuous yield) + key sensitivity relationship

Definition (duration): with bond price $B = \sum_i c_i e^{-yt_i}$,

$$D = \frac{\sum_{i=1}^n t_i c_i e^{-yt_i}}{B} = \sum_{i=1}^n t_i \cdot \underbrace{\frac{c_i e^{-yt_i}}{B}}_{\text{PV weight}}.$$

Interpretation: D is the *PV-weighted average time* you wait to receive value.

Key first-order approximation:

$$\Delta B \approx -B D \Delta y \quad \Longleftrightarrow \quad \frac{\Delta B}{B} \approx -D \Delta y.$$

So bigger $D \Rightarrow$ more interest-rate (yield) sensitivity.

Practice A: (Hull 4.18) Bond price & duration

A 5-year bond with a yield of 7% (continuously compounded) pays an 8% coupon at the end of each year. Assume face value is \$100.

(a) What is the bond's price?

(b) What is the bond's duration?

TA Key for Practice A

Cash flows: $c_1 = \dots = c_4 = 8$, $c_5 = 108$.

Price:

$$B = \sum_{t=1}^4 8e^{-0.07t} + 108e^{-0.07 \cdot 5} \approx 103.05.$$

Duration:

$$D = \frac{\sum_{t=1}^4 t \cdot 8e^{-0.07t} + 5 \cdot 108e^{-0.35}}{B} \approx \frac{445.54}{103.05} \approx 4.32 \text{ years.}$$

2 Part 2: Forward Rates & Forward Pricing

Module B) Forward rates & forward pricing (investment asset w/o income)

Forward rate from spot rates (cont. comp.): the implied forward rate for $[T_1, T_2]$ is

$$f_0(T_1, T_2) = \frac{R(T_2)T_2 - R(T_1)T_1}{T_2 - T_1}.$$

Intuition: It is the rate locked in *today* for a future period $[T_1, T_2]$, acting as the “bridge” between the shorter and longer spot rates. **Using the spot rate today to calculate the time-value of future time periods.**

Forward price (investment asset, no income, no storage costs):

$$F_0(T) = S_0 e^{rT}.$$

Intuition (cost of carry): buying now costs S_0 and financing to T costs a factor e^{rT} .

Arbitrage logic (static):

- If $F_0(T) > S_0 e^{rT}$: *cash-and-carry* $\Rightarrow$ borrow S_0 , buy asset, short forward.
- If $F_0(T) < S_0 e^{rT}$: *reverse cash-and-carry* $\Rightarrow$ short asset, invest proceeds, long forward.

Practice B: (Hull 5.2) Forward price on a non-dividend-paying stock

Suppose that you enter into a 6-month forward contract on a non-dividend-paying stock when the stock price is \$30 and the risk-free interest rate (with continuous compounding) is 5% per annum. What is the forward price?

TA Key for Practice B

No income, so use base formula:

$$F_0 = S_0 e^{rT} = 30 e^{0.05 \cdot 0.5} = 30 e^{0.025} \approx 30.76.$$